Ongoing Work on Contextual Neural Machine Translation

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Motivation

- **Most neural machine translation (NMT) systems translate sentence by sentence**
- **Make use of surrounding sentences to exploit context information**
  - **reduce ambiguities in translation choices**
  - **example: cross-sentence anaphora in English→German**

Source sentences	**Correct translation of 'it'**
The monkey ate the banana. Because it was hungry.	**er**
The monkey ate the banana. Because it ripe.	**sie**
The monkey ate the banana. Because it was tea-time.	**es**

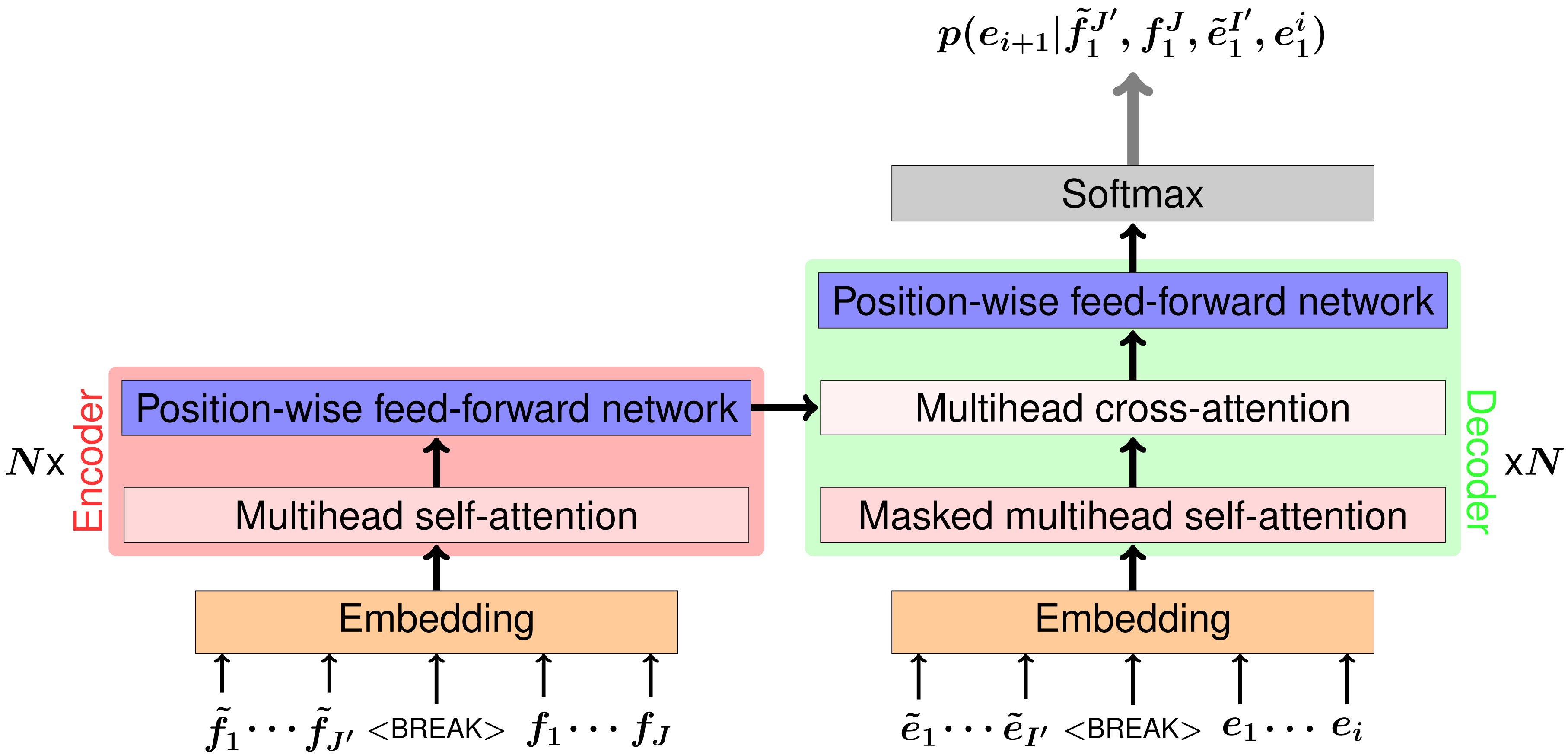
State of the Art Literature

- Single encoder: concatenate sentences on source-/target side [Tiedemann & Scherrer 17, Agrawal & Turchi^{+} 18]

- Multi-Encoder with attention mechanism
  - Feed surrounding sentences into seperate encoders and combine context vectors:
    - concatenation (like in [Zoph & Knight 16])
    - gated sum [Bawden & Sennrich^{+} 18]
    - hierarchical attention [Bawden & Sennrich^{+} 18]

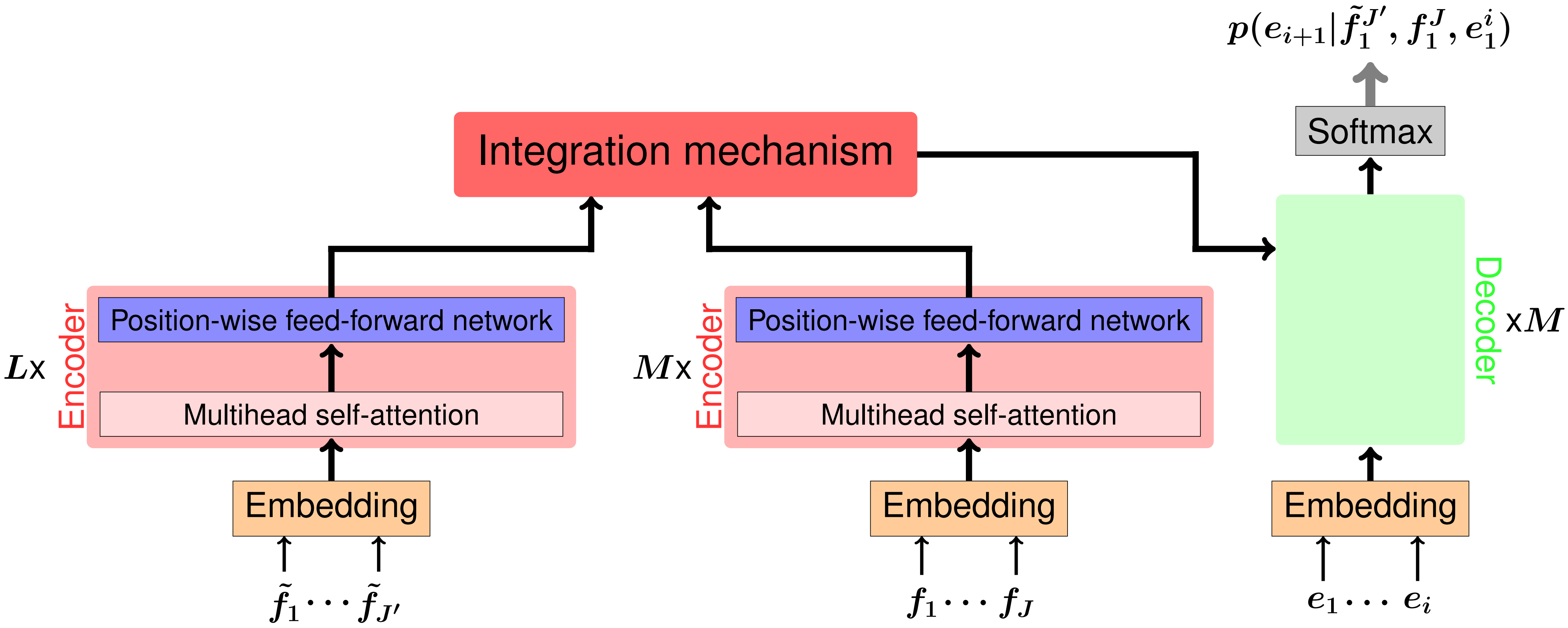
Remark: above methods only consider *local* context

Transformer Architecture with Concatenated Context Sentences



- **one previous context-sentence on the source ($\tilde{f}_1^{J'}$) and target ($\tilde{e}_1^{I'}$) side**
- **$<$BREAK$>$ token seperates the sentences**

Multi-encoder Transformer Architecture



- Here: one source-context sentence $\tilde{f}_1^{J'}$ is encoded seperately
- Integration mechanism is unclear: concatenation, gating mechanism, . . .

Ongoing Work at i6

1. Reproduce results from [Agrawal & Turchi$^+$ 18]
   - Transformer model
   - IWSLT 2017 English$\rightarrow$Italian
   - concatenate sentences with special token $<$BREAK$>$, example: `sentence_1` $<$BREAK$>$ `sentence_2`
2. Compress context-sentences by dropping words
   - first n most-common words in training corpus
   - pre-defined stopword list
3. Test method of concatenation ($+$ word dropping) on WMT 2018 German$\rightarrow$English data as well

Concatenation

(a) Single target sentence.

Source	$\rightarrow$	Target
1 previous + current	$\rightarrow$	current
2 previous + current	$\rightarrow$	current
1 previous + current + 1 next	$\rightarrow$	current
current	$\rightarrow$	current

(b) Multiple target sentences.

Source	$\rightarrow$	Target
1 previous + current	$\rightarrow$	1 previous + current
1 previous + current + 1 next	$\rightarrow$	1 previous + current

Dropping Words in Context Sentences

- Remove punctuations first
- Drop $MC \in \mathbb{N}$ most common words from the training corpus
  - $DW \equiv$ Percentage of dropped words in a corpus
  - For $MC = 90$ we obtain $DW = 57\%$ on IWSLT En→It data

And that led us directly to the Olympic Village .

$\downarrow$

led directly Olympic Village

- Concatenate shortened context-sentences to current one that is untouched

Data Statistics

		IWSLT'17 EN-IT		WMT'18 DE-EN	
		EN	IT	DE	EN
Train.	Sentences	232K		5.9M	
	Running words	4.7M	4.4M	160.9M	157.7M
Dev.		*tst2010*		*newstest2016*	
	Sentences	1.5K		3K	
	Running words	31K	29K	77K	73K
Test		*tst2017*		*newstest2017*	
	Sentences	1.1K		3K	
	Running words	22K	20K	76K	73K

Training

- Transformer model [Vaswani & Shazeer^{+} 17] with 6 encoder/decoder layers
- Model dimensionality: 512
- Feed-forward network dimensionality: 2048
- Attention-heads: 8

- Adam optimizer [Kingma & Ba 14]
  - learning rate: 0.0002
  - 2,000 warmup steps

- 32,000 Byte-pair encoding merges [Sennrich & Haddow^{+} 16]

Results: IWSLT 2017 English→Italian

Concatenation baselines.

			tst2017 BLEU [%]	TER [%]
current	→	current	29.89	52.0
1 previous + current	→	current	30.02	51.8
2 previous + current	→	current	30.00	52.1
1 previous + current + 1 next	→	current	30.46	52.0
current + 1 next	→	current	29.93	51.9
1 previous + current	→	1 previous + current	30.64	51.0
1 previous + current + 1 next	→	1 previous + current	30.64	51.4

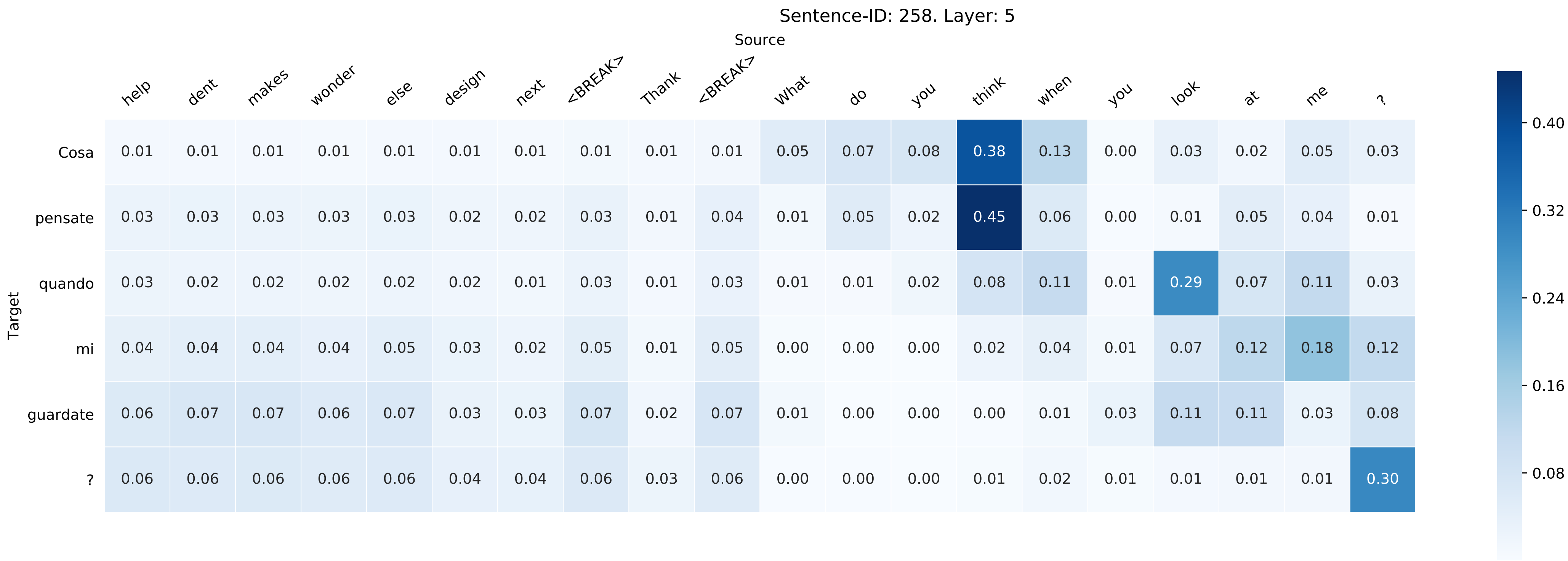
Dropping most common words experiments: *single* target sentence.

Source	→ Target	MC	DW	*tst2017* BLEU [%]	TER [%]
current	→ current	0	0%	29.89	52.0
1 previous + current	→ current	0	0%	30.02	51.8
		100	59%	30.03	52.0
2 previous + current	→ current	0	0%	30.00	52.1
		90	57%	30.35	51.4
1 previous + current + 1 next	→ current	0	0%	30.46	52.0
		90	57%	30.14	51.8
current + 1 next	→ current	0	0%	29.93	51.9
		100	59%	30.00	51.6

Dropping most common words experiments: *multiple* target sentences.

Source	→	Target	MC	DW	*tst2017* BLEU [%]	TER [%]
1 previous + current	→	1 previous + current	0	0%	30.64	51.0
			90	57%	30.09	52.0
1 previous + current + 1 next	→	1 previous + current	0	0%	30.64	51.4
			150	64%	30.18	51.6

IWSLT English→Italian: Cross-Attention Examples



Sentence-ID: 238. Layer: 3

Target \ Source	homes	designed	shared	ground	up	<BREAK>	look	<BREAK>	What	if	cities	em@@	br@@	ac@@	ed	a	culture	of	sharing	?
E	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.70	0.25	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
se	0.05	0.08	0.08	0.07	0.08	0.10	0.05	0.12	0.02	0.30	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
le	0.01	0.02	0.02	0.01	0.02	0.02	0.01	0.02	0.01	0.01	0.69	0.13	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.00
città	0.03	0.04	0.04	0.04	0.05	0.06	0.02	0.06	0.00	0.00	0.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
avessero	0.02	0.02	0.02	0.02	0.02	0.03	0.01	0.03	0.00	0.00	0.02	0.33	0.14	0.02	0.20	0.12	0.00	0.00	0.00	0.00
abbracci@@	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.23	0.33	0.07	0.21	0.13	0.00	0.00	0.00	0.00
ato	0.07	0.16	0.13	0.14	0.15	0.17	0.04	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00
una	0.02	0.03	0.02	0.03	0.03	0.03	0.02	0.04	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.49	0.21	0.03	0.01	0.01
cultura	0.01	0.02	0.02	0.02	0.02	0.02	0.01	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.75	0.08	0.00	0.00
della	0.01	0.02	0.02	0.02	0.02	0.03	0.01	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.36	0.46	0.00
condivisione	0.02	0.04	0.03	0.04	0.04	0.05	0.01	0.03	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.71	0.00
?	0.02	0.02	0.02	0.02	0.02	0.02	0.01	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.13	0.28

Results: WMT 2018 German→English

Concatenation baselines.

			newstest2017 BLEU [%]	*newstest2017* TER [%]
current	→	current	30.36	51.0
1 previous + current	→	current	30.13	51.4
2 previous + current	→	current	30.32	51.2
1 previous + current + 1 next	→	current	30.28	51.2
current + 1 next	→	current	30.15	51.0
1 previous + current	→	1 previous + current	30.10	51.2
1 previous + current + 1 next	→	1 previous + current	30.18	51.2

Dropping most common words experiments.

Source	→ Target	MC	DW	newstest2017 BLEU [%]	newstest2017 TER [%]
current →	current	0	0%	30.36	51.0
1 previous + current →	current	0	0%	30.13	51.4
		300	57%	29.98	50.86
2 previous + current →	current	0	0%	30.32	51.2
		300	57%	27.66	60.08
1 previous + current + 1 next →	current	0	0%	30.28	51.2
		300	57%	30.06	51.8
current + 1 next →	current	0	0%	30.15	51.0
		300	57%	30.20	51.2
1 previous + current + 1 next →	1 previous + current	0	0%	30.18	51.4
		300	57%	27.31	54.4

Summary

- Concatenation of sentences yield slight performance increases with regards to BLEU and TER metrics
  - IWSLT En→It: up to +0.75 BLEU improvement
  - WMT De→En: similar performance compared to baseline model
- Simple compression by dropping most common words
  - IWSLT En→It: similar performance compared to concatenated models
  - WMT De→En: mostly degradation in BLEU/TER
- English language is easier in terms of linguistic properties
- Dropping stopwords does not yield improvements

Future Work

- Implementation of multi-encoder approaches [Bawden & Sennrich+ 18, Voita & Sennrich+ 18]
- Consider global context information: e.g. topic embedding
- Test approaches in discourse-heavy data sets like OpenSubtitles [Lison & Tiedemann 16]
- More sophisticated evaluation with regards to discourse-phenomena
  - pronoun translation
  - co-reference

Thank you for your attention

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Appendix: Results from [Agrawal & Turchi^{+} 18]

- Transformer model with concatenation
- IWSLT 2017 English→Italian

			tst2017	
			BLEU [%]	TER [%]
current	→	current	29.2	52.8
1 previous + current	→	current	29.4	52.5
2 previous + current	→	current	29.8	51.9
1 previous + current + 1 next	→	current	30.6	51.1
current + 1 next	→	current	29.7	51.9
1 previous + current	→	1 previous + current	29.5	52.1
1 previous + current + 1 next	→	1 previous + current	31.5	49.7